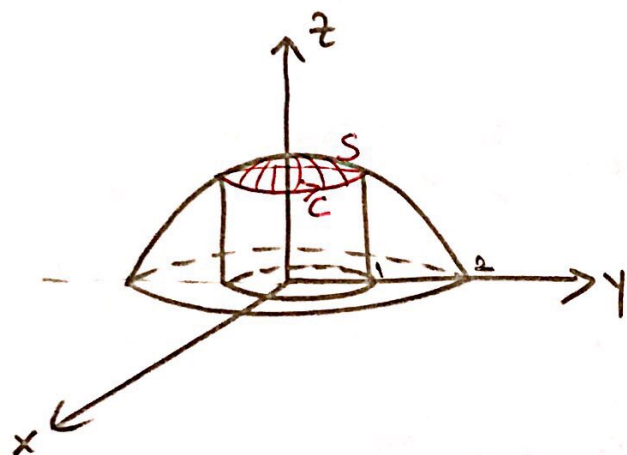


⊗ Use Stoke's Theorem to compute $\iint_S \text{Rot} \vec{F} \cdot \vec{N} ds$ where $\vec{F} = xz\vec{i} + yz\vec{j} + xy\vec{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.



⊗ To find the boundary curve C ;

we solve: $x^2 + y^2 + z^2 = 4$
and $x^2 + y^2 = 1$

So, C is the circle given by:

$$C: \underline{x^2 + y^2 = 1, \quad z = \sqrt{3}}$$

$$I = \iint_S \text{Rot} \vec{F} \cdot \vec{N} ds = \int_C \vec{F} \cdot d\vec{r}$$

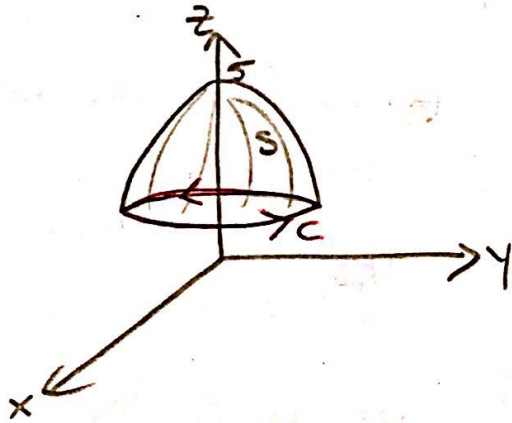
$$I = \int_C xz dx + yz dy + xy dz$$

$$C: \begin{cases} x = \cos \theta \\ y = \sin \theta \\ z = \sqrt{3} \end{cases} \quad \begin{cases} dx = -\sin \theta d\theta \\ dy = \cos \theta d\theta \\ dz = 0 \end{cases}$$

$$I = \int_0^{2\pi} \sqrt{3} \cos \theta (-\sin \theta) d\theta + \sqrt{3} \sin \theta (\cos \theta) d\theta + 0$$

$$I = \sqrt{3} \int_0^{2\pi} 0 dt \Rightarrow I = 0$$

⊛ Use Stoke's Theorem to evaluate $\iint_S \text{Rot } \vec{F} \cdot \vec{N} ds$ where $\vec{F} = z^2 \vec{i} - 3xy \vec{j} + x^3y^3 \vec{k}$ and S is the part of $z = 5 - x^2 - y^2$ above the plane $z = 1$.



In this case the boundary curve C will be where the surface intersects the plane $z = 1$ and so will be the curve $x^2 + y^2 = 4$ at $z = 1$.

$$\iint_S \text{Rot } \vec{F} \cdot \vec{N} ds = \int_C \vec{F} \cdot d\vec{r}$$

$$C: \begin{matrix} x^2 + y^2 = 4 \\ z = 1 \end{matrix} \Rightarrow \begin{matrix} x = 2\cos\theta \\ y = 2\sin\theta \\ z = 1 \end{matrix} \left\{ \begin{matrix} dx = -2\sin\theta d\theta \\ dy = 2\cos\theta d\theta \\ dz = 0 \end{matrix} \right.$$

$$I = \int_C \vec{F} \cdot d\vec{r} = \int_C z^2 dx - 3xy dy + x^3y^3 dz$$

$$I = \int_0^{2\pi} 1 \cdot (-2\sin\theta) d\theta - 3 \cdot (2\cos\theta)(2\sin\theta) 2\cos\theta d\theta + 0$$

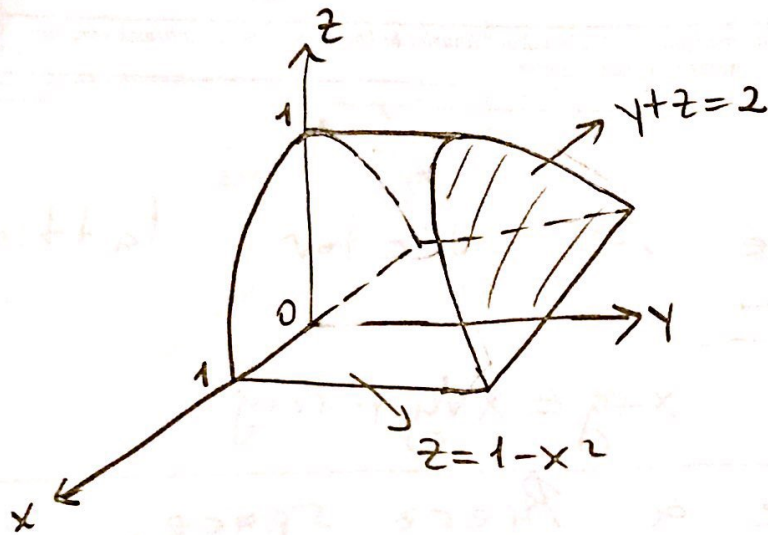
$$I = \int_0^{2\pi} (-2\sin\theta - 24\sin\theta \cos^2\theta) d\theta$$

$$I = \int_0^{2\pi} -2\sin\theta d\theta - \int_0^{2\pi} 24\sin\theta \cos^2\theta d\theta \quad \left\{ \begin{matrix} \cos\theta = t \\ -\sin\theta d\theta = dt \end{matrix} \right.$$

$$I = 2\cos\theta \Big|_0^{2\pi} + 8\cos^3\theta \Big|_0^{2\pi}$$

$$I = 0$$

(*) Compute $I = \iiint_S \vec{F} \cdot \vec{N} ds$ for the vector field $\vec{F} = xy\vec{i} + (y^2 + e^{xz^2})\vec{j} + \sin(xy)\vec{k}$ where S is the surface of the region E bounded by the parabolic cylinder $z = 1 - x^2$ and the planes $z = 0, y = 0, y + z = 2$.



It would be extremely difficult to evaluate the given surface integral directly. We would have to evaluate 4 surface integrals to 4 pieces of S .

Also, the divergence of $\vec{F}$ is much less complicated than $\vec{F}$ itself;

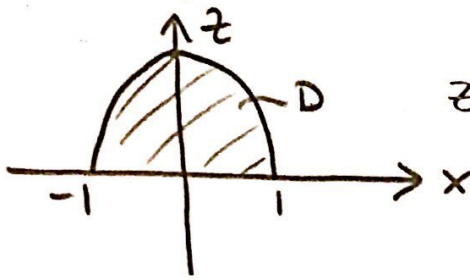
$$\operatorname{div} \vec{F} = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(y^2 + e^{xz^2}) + \frac{\partial}{\partial z}(\sin(xy))$$

$$\operatorname{div} \vec{F} = y + 2y = 3y$$

So, we use the Divergence Theorem to transform the given surface integral into a triple integral.

$$I = \iint_S \vec{F} \cdot \vec{N} d\vec{b} = \iiint_V \text{div } \vec{F} dV$$

Have projection on xz -plane;



$$z = 1 - x^2, \quad y_1 = 0 \\ y_2 = 2 - z$$

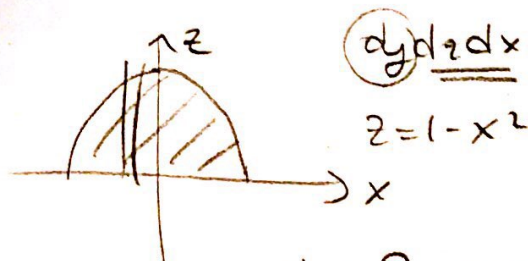
$$I = \iiint_V 3y dV = \iint_D \int_{y_1}^{y_2} 3y dy dz dx$$

$$I = \iint_D \int_0^{2-z} 3y dy dz dx = 3 \int_{-1}^1 \int_0^{1-x^2} \int_0^{2-z} y dy dz dx$$

$$I = \frac{3}{2} \int_{-1}^1 \int_0^{1-x^2} y^2 \Big|_0^{2-z} dz dx = \frac{3}{2} \int_{-1}^1 \int_0^{1-x^2} (2-z)^2 dz dx$$

$$I = \frac{3}{2} \int_{-1}^1 \left[-\frac{(2-z)^3}{3} \right]_0^{1-x^2} dx = -\frac{1}{2} \int_{-1}^1 [(x^2+1)^3 - 8] dx$$

$$I = -\frac{1}{2} \int_{-1}^1 (x^6 + 3x^4 + 3x^2 + 1 - 8) dx = \frac{184}{35}$$



$$\underline{\underline{dy dz dx}}$$

$$z = 1 - x^2$$

$$y_1 = 0 \quad y_2 = 2 - z$$

$$I = \iiint_D \dots dy dz dx$$

$$I = \int_{-1}^1 \int_0^{1-x^2} \int_0^{2-z} \dots dy dz dx$$